§1.564 tabulated_is_entire_iff_left_cover

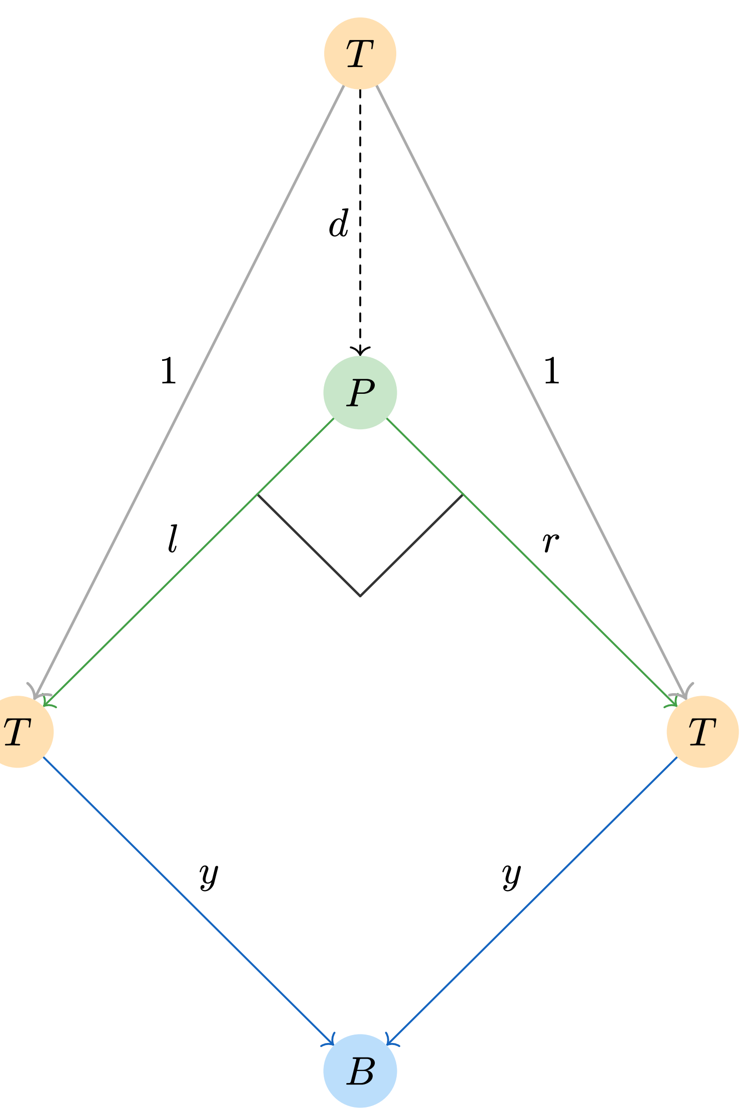

Figure 1: Pullback of (y, y). The pair $\langle 1_T, 1_T\rangle$ lifts uniquely to $d : T \to P$.

d unique: $dl = 1_T$,
$dr = 1_T$
with $ly = ry$

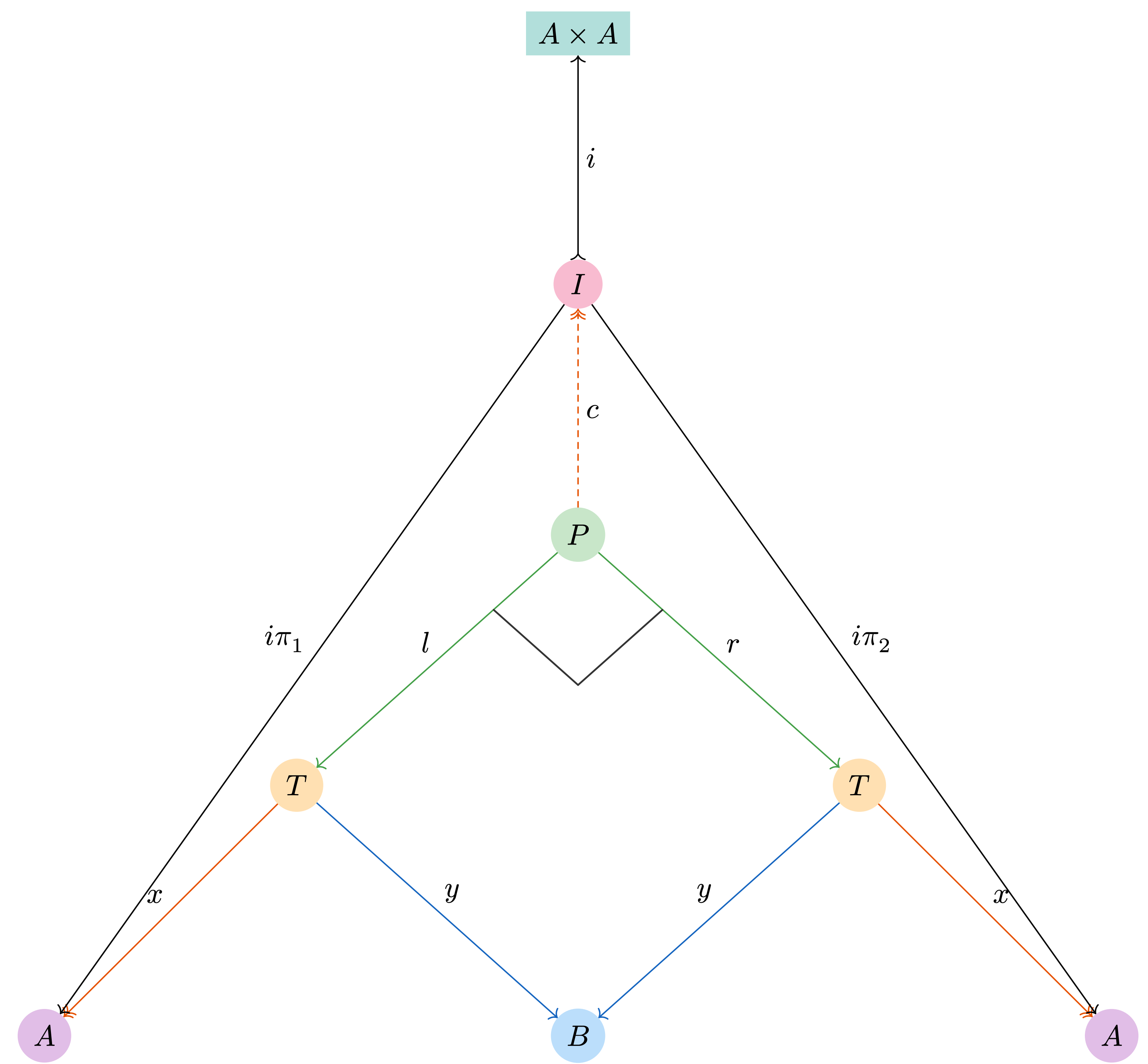

Figure 2: Composition $\mathbb{R}^{\circ}$: pullback of (y, y) then image into $A \times A$.

$s := \langle lx, rx\rangle : P \to A \times A$ (unique product pairing)
$ci = s$, c cover (unique: i monic), i monic

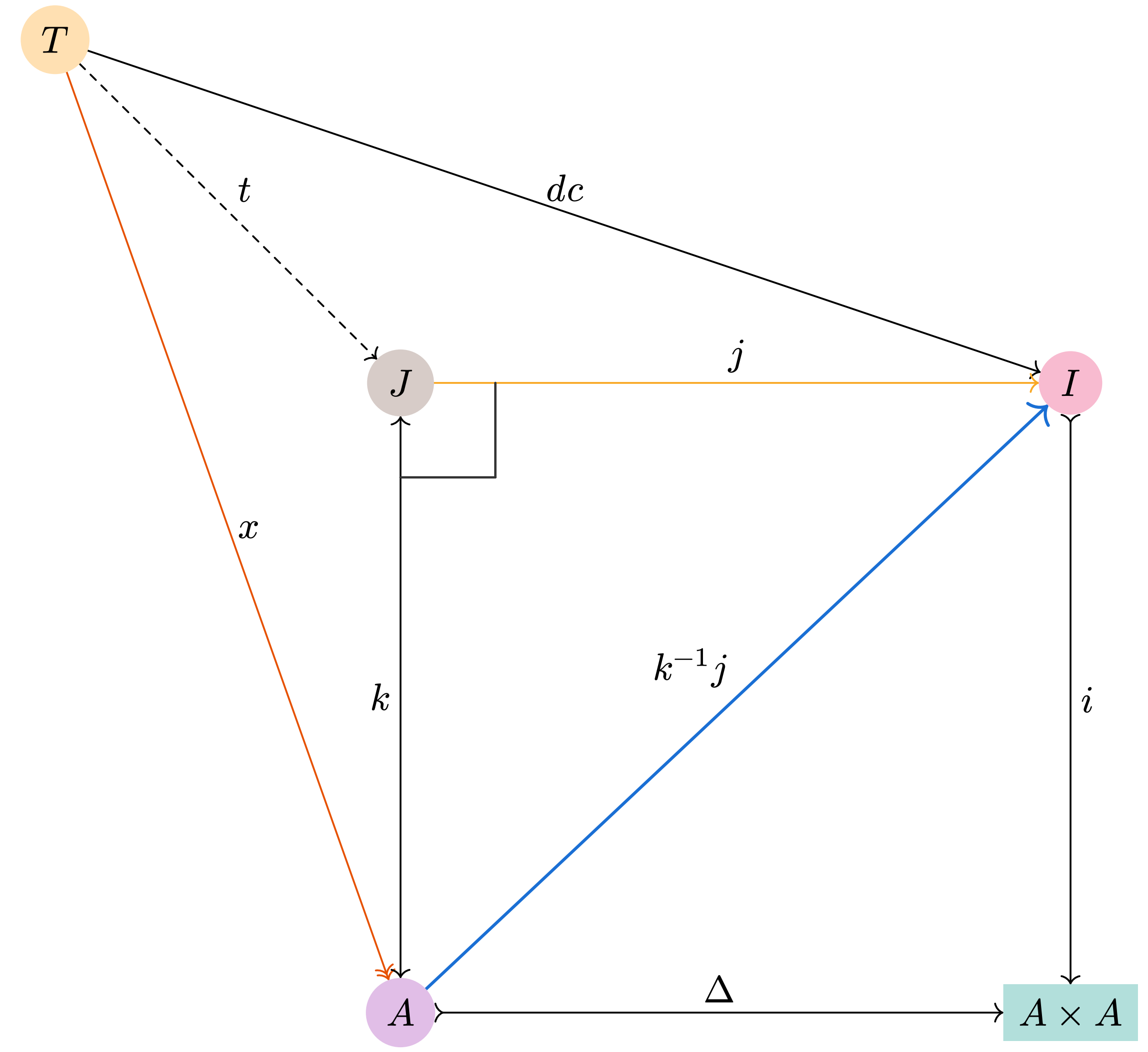

Figure 3: Pullback J of (Δ, i). $x = tk$ cover with k monic $\Rightarrow$ k iso (§1.363). $h := k^{-1}\ j$ witnesses $1 \leq \mathbb{R}^{\circ}$.

$J :=$ pullback of (Δ, i),
t unique: $tk = x$, $tj = dc$
$(dc)i = x\Delta$
$h(i\pi_1) = 1$,
$h(i\pi_2) = 1$

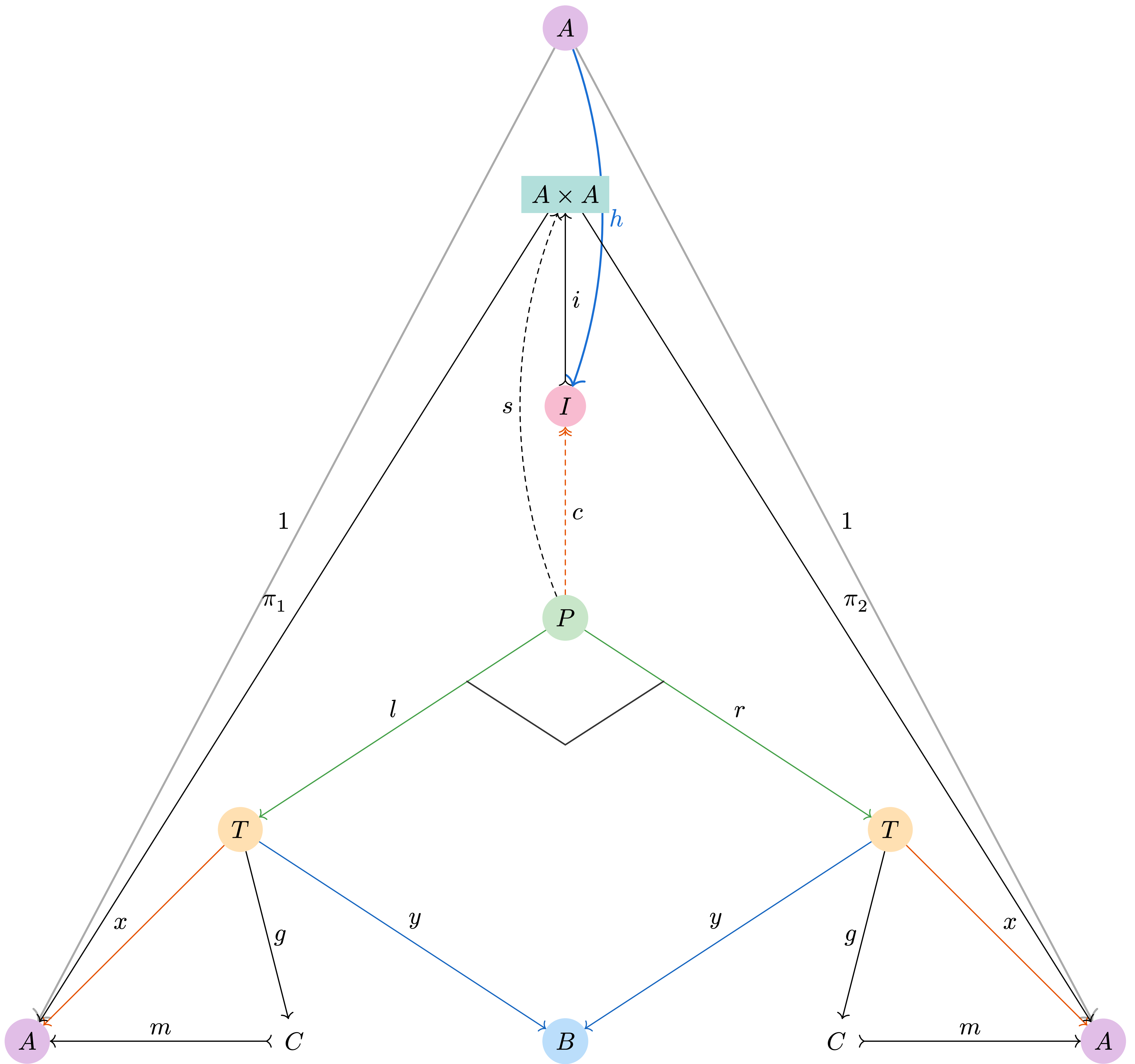

Figure 4: Entire condition $1_A \leq \mathbb{R}^{\circ}$ and factorization $x = gm$ with m monic.

$1_A \leq \mathbb{R}^{\circ}$: $h(i\pi_1) = 1_A$,
$h(i\pi_2) = 1_A$,
$hi = \Delta = \langle 1, 1\rangle$
$s := \langle lx, rx\rangle$ (unique),
$ci = s$ (c unique: i monic)
$x = gm$,
$m : C \to A$ monic

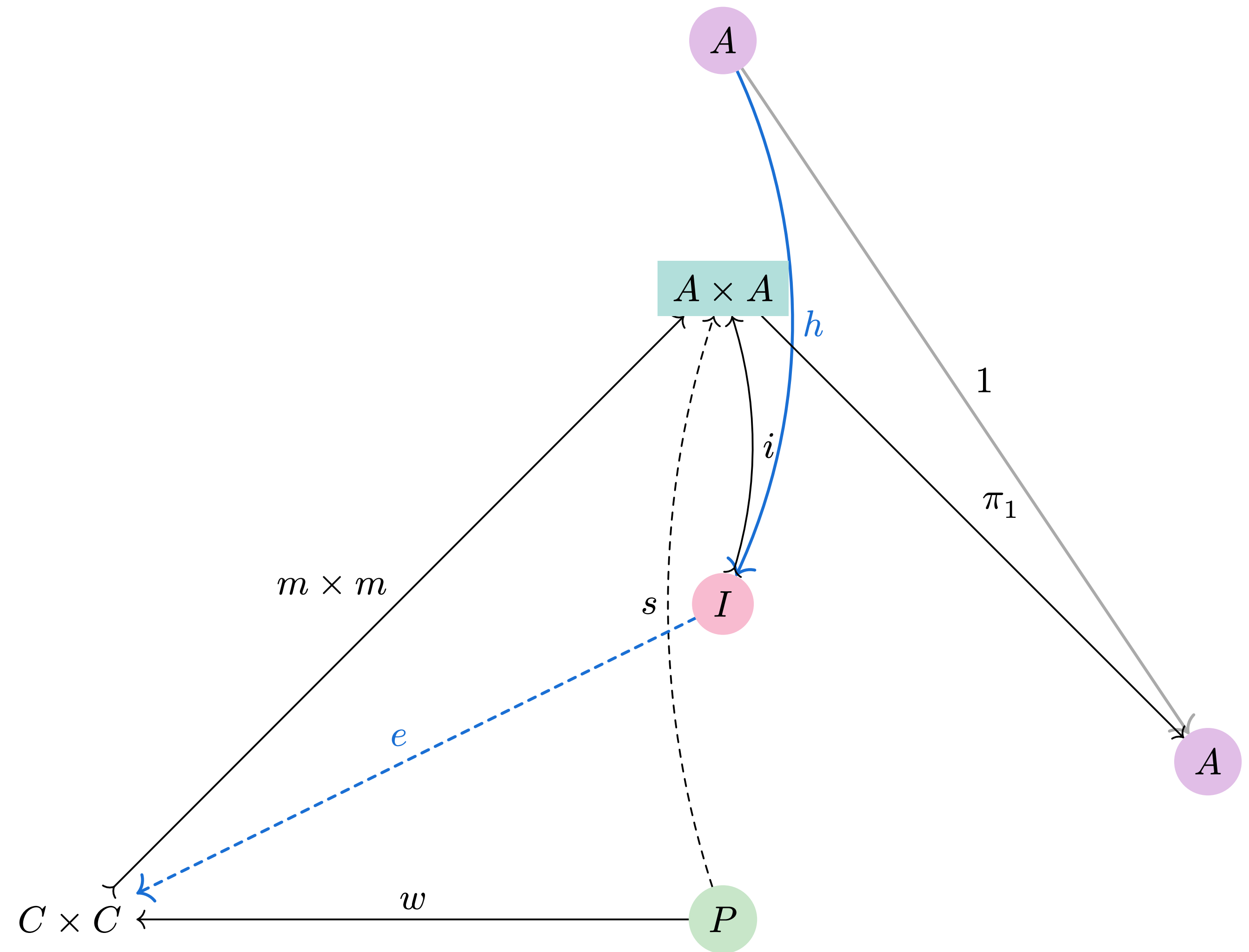

Figure 5: Image minimality lifts I into $C \times C$, producing e with $e(m \times m) = i$. Then $(he)\pi_1$ left-inverts m.

$w := \langle lg, rg\rangle$, $w(m \times m) = s$, $m \times m$ monic
I = smallest monic subobject s factors through
$\Rightarrow \exists! e : I \to C \times C$ with $e(m \times m) = i$

$((he)\pi_1)m = (he)(\pi_1 m) = (he)((m \times m)\pi_1) = (hi)\pi_1 = 1_A$
(① $\pi_1 m = (m \times m)\pi_1$;
② $e(m \times m) = i$;
③ $hi\pi_1 = 1_A$)
m monic + left inverse $\Rightarrow$ iso $\Rightarrow$ $x = gm$ cover

Legend:
->> hollow head = cover,
>-> tailed = monic,
double arrow = iso,
dashed = induced / witness,
blue = witness,
⌝ = pullback corner.